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Course Title: Simulation and Modeling lab

Lab Assignment 2

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Unique Matrix Value Investigation

Objective

To analyze how changing matrix values affects matrix properties (shape, determinant, rank, eigenvalues, and invertibility), using a matrix generated uniquely from the student's own ID.

Step 1: Personal Matrix Construction

From the student ID 0432320005101132, the last two digits are extracted as follows:

- $d1$ = last digit of ID = 2
- $d2$ = second last digit of ID = 3

Substituting into the given matrix form:

$$A = \begin{bmatrix} d1+2 & d2+1 \\ 2d1 & d2+2 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 4 & 5 \end{bmatrix}$$

Step 2: Analysis of Matrix A

Matrix A was analyzed using NumPy. The computed properties are:

- Shape: (2, 2)

- Determinant: 4.0
- Rank: 2 (full rank, since the determinant is non-zero)
- Eigenvalues: 0.4689 and 8.5311
- Inverse (exists since $\det \neq 0$): $\begin{bmatrix} 1.25 & -1 \\ -1 & 1 \end{bmatrix}$

Step 3: Value Change Experiment

A new matrix B was created by adding +1 to exactly one element of A. The element changed is the top-left element, $A[0][0]$ (originally 4, changed to 5):

$$B = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$$

Step 4: Analysis of Matrix B

- Shape: (2, 2)
- Determinant: 9.0
- Rank: 2 (full rank, since the determinant is non-zero)
- Eigenvalues: 1.0 and 9.0
- Inverse (exists since $\det \neq 0$): $\begin{bmatrix} 0.5556 & -0.4444 \\ -0.4444 & 0.5556 \end{bmatrix}$

Python Code

```
import numpy as np

#Step 1: Build the personal matrix from Student ID
student_id = "0432320005101132"

d1 = int(student_id[-1])    # last digit -> 2
d2 = int(student_id[-2])    # second last digit -> 3

A = np.array([
    [d1 + 2, d2 + 1],
    [2 * d1, d2 + 2]
], dtype=float)

print("Matrix A:\n", A)

#Step 2: Analyze Matrix A
print("\nAnalysis of Matrix A ")
print("Shape of A      :", A.shape)
print("Determinant of A :", np.linalg.det(A))
print("Rank of A       :", np.linalg.matrix_rank(A))
print("Eigenvalues of A :", np.linalg.eigvals(A))

if np.linalg.det(A) != 0:
    print("Inverse of A:\n", np.linalg.inv(A))
```

```

#Step 3: Value Change Experiment
# Change top-left element by +1
B = A.copy()
B[0, 0] += 1

print("\nMatrix B:\n", B)

#Step 4: Re-analyze Matrix B
print("\nAnalysis of Matrix B ")
print("Shape of B      :", B.shape)
print("Determinant of B :", np.linalg.det(B))
print("Rank of B       :", np.linalg.matrix_rank(B))
print("Eigenvalues of B :", np.linalg.eigvals(B))

if np.linalg.det(B) != 0:
    print("Inverse of B:\n", np.linalg.inv(B))

```

Program Output

```

print("inverse of B:\n", np.linalg.inv(B))

```

Matrix A:
[[4. 4.]
[4. 5.]]

Analysis of Matrix A
Shape of A : (2, 2)
Determinant of A : 4.0
Rank of A : 2
Eigenvalues of A : [0.46887113 8.53112887]
Inverse of A:
[[1.25 -1.]
[-1. 1.]]

Matrix B:
[[5. 4.]
[4. 5.]]

Analysis of Matrix B
Shape of B : (2, 2)
Determinant of B : 8.999999999999999
Rank of B : 2
Eigenvalues of B : [9. 1.]
Inverse of B:
[[0.55555556 -0.44444444]
[-0.44444444 0.55555556]]

Property Comparison: Matrix A vs Matrix B

Property	Matrix A	Matrix B
Shape	(2, 2)	(2, 2)
Determinant	4.0	9.0
Rank	2	2
Eigenvalues	0.4689, 8.5311	1.0, 9.0
Inverse	[[1.25, -1], [-1, 1]]	[[0.5556, -0.4444], [-0.4444, 0.5556]]

Step 5: Comparison & Explanation

1. How did the determinant change and why?

The determinant increased from 4.0 (Matrix A) to 9.0 (Matrix B). Since $\det(A) = (d1+2)(d2+2) - (d2+1)(2d1)$, increasing the top-left element by 1 directly increases the first product term in the determinant formula while the second term stays fixed, which raises the overall determinant value. This shows that the determinant is sensitive to changes in individual elements, and the effect depends on which element is changed and where it sits in the determinant formula.

1. Did the rank change? Explain.

No, the rank did not change — it remained 2 for both matrices. This is because both A and B are 2x2 matrices with non-zero determinants (4.0 and 9.0 respectively), meaning both matrices are still full rank (linearly independent rows/columns). A small change of +1 to one element was not enough to make the rows of the matrix linearly dependent, so both matrices retained full rank.

2. How did the eigenvalues respond to the value change?

The eigenvalues shifted from approximately 0.4689 and 8.5311 (Matrix A) to exactly 1.0 and 9.0 (Matrix B). Both eigenvalues increased, and notably the eigenvalues of B are close to round numbers. Since the sum of the eigenvalues equals the trace of the matrix (which increased from 9 to 10 due to the +1 change on the diagonal) and their product equals the determinant (which increased from 4 to 9), both eigenvalues increased in response to the increase in the trace and determinant.

3. Is B easier or harder to invert than A? Why?

Matrix B is comparatively easier/more stable to invert than Matrix A. A matrix becomes harder to invert as its determinant approaches zero (since the inverse involves dividing by the determinant, and inversion becomes numerically unstable or impossible as $\det \rightarrow 0$). Since Matrix B has a larger determinant (9.0) than Matrix A (4.0), and its eigenvalues (1.0 and 9.0) are further from zero than A's smallest eigenvalue (0.4689), Matrix B is better conditioned and therefore easier and more numerically stable to invert.

Conclusion

This investigation demonstrates that changing a single matrix element can significantly affect several matrix properties. While the shape and rank remained unchanged, the determinant, eigenvalues, and inverse matrix changed noticeably. The experiment highlights how sensitive matrix characteristics are to value modifications and reinforces the importance of determinants and eigenvalues in understanding matrix behavior.

